also cases where additional levels of refinement are required, so your data design could extend to Diamond and Platinum layers as well. The Medallion architecture is a good fit for data science projects. By selecting, cleaning, and organizing data for a specific purpose, curated data can help improve the data's accuracy, relevance, and usability.

> **Note**
>
> This book will introduce quite a few project tasks, such as *building curated datasets*, that often fall under the domain of a Data Engineer. There will also be tasks that Machine Learning Engineers, data analysts, and so on commonly perform. We include all this work in our examples because roles are blurred in today's fast-moving world and will vary by company. Titles and expectations quickly evolve, and it's imperative to have a handle on the entire data lifecycle.

Maintaining a Bronze layer allows us to go back to the original data when we want to create a different feature, solve a new problem that requires us to look at historical data from another point of view, or simply maintain the raw level of our data for governance purposes. As the world of technology evolves, it's crucial to stay current and follow trends, but the core principles remain constant. Each of the following sections in this chapter (*Transforming data to Delta with Auto Loader*, *Delta Live Tables for the Bronze layer*, and *Maintaining and optimizing Delta Tables*) will cover DI platform features that facilitate building the Bronze layer. In the *Applying our learning* section, we apply the features to our example projects.

Transforming data to Delta with Auto Loader

Before we can transform our raw data, we must first ingest it. Auto Loader is a feature that incrementally and efficiently processes new data files as they arrive in cloud storage. You can use Auto Loader to automate your data ingestion process, significantly enhancing your data product's workflow efficiency. It can ingest data from cloud storage and streaming data sources and be configured to run on a schedule or be triggered manually.

Here are some benefits of using Databricks Auto Loader:

- **Reduces overhead**: Auto Loader maintains checkpoints, removing the need to track which data is new. Auto Loader handles that all on its own.
- **Improves data quality**: Auto Loader can automatically detect schema changes and rescue any new data columns, so you can be confident that your data is accurate.
- **Increases data agility**: Auto Loader can help you quickly and easily ingest new data sources so you can be more agile in responding to changes in your business.
- **Flexible ingestion**: Auto Loader can stream files in batches or continuously. Meaning, it can consume batch data as a stream to reduce the overhead of a more manual batch pipeline.